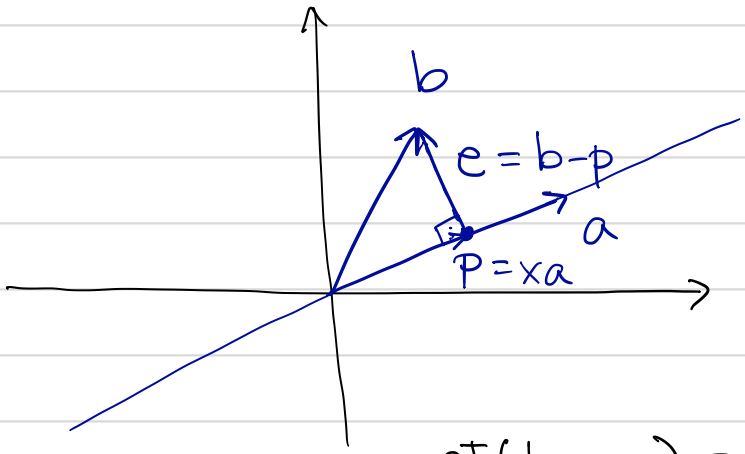
Lecture 23: Projections, Least squares



$$a^T(b - xa) = 0$$

$$a^Ta \cdot x = a^Tb$$

$$\boxed{x = \frac{a^Tb}{a^Ta}} \qquad \boxed{p = a\frac{a^Tb}{a^Ta}}$$

$$= P \cdot b$$

$$\boxed{\text{Projection Matrix: } P = \frac{aa^T}{a^Ta}}$$

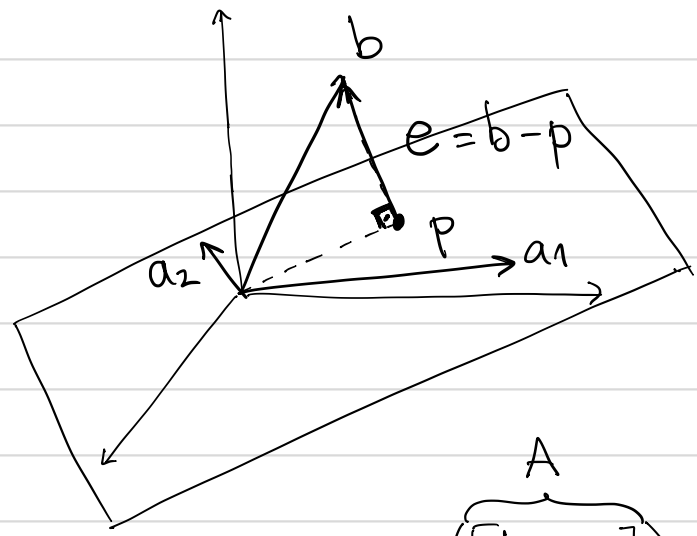
$\text{col}(P)$ = line through a
$= \text{span}(a)$

$\text{rank}(P) = 1$

$\boxed{\begin{matrix} P^T = P \\ P^2 = P \end{matrix}}$ (P is symmetric)

Why project?
Because $Ax = b$ may have no solutions.

Solve $A\hat{x} = p$
↳ Projection of b onto $\text{col}(A)$.



Plane of a_1, a_2 $= \text{col}\left(\overbrace{\begin{bmatrix} | & | \\ a_1 & a_2 \\ | & | \end{bmatrix}}^{A}\right)$

$$p = \hat{x}_1 a_1 + \hat{x}_2 a_2 = A\hat{x}$$

Find: $\hat{x}$

Key: $e = b - A\hat{x}$ is perpendicular to the plane

$$a_1^T(b - A\hat{x}) = 0 \qquad a_2^T(b - A\hat{x}) = 0$$

$$\underbrace{\begin{bmatrix} - a_1^T - \\ - a_2^T - \end{bmatrix}}_{A^T} (b - A\hat{x}) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$A^T(b - A\hat{x}) = 0$$

$$\boxed{A^T A \hat{x} = A^T b}$$

$$\hat{x} = (A^T A)^{-1} A^T b$$

1D: $\frac{aa^T}{a^Ta}$

$$p = A\hat{x} = A(A^TA)^{-1}A^T b$$

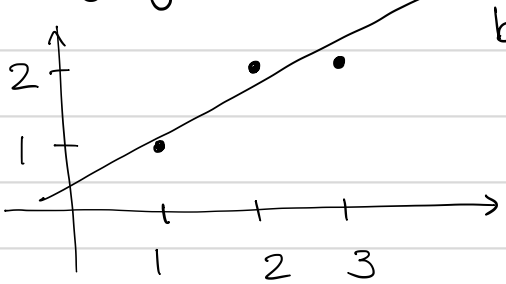
Projection matrix: $P = A(A^TA)^{-1}A^T$

- $\cancel{A\, A^{-1} (A^T)^{-1} \cdot A^T = I}$ $\quad P^T = P, \quad P^2 = P$
- $P^T = P$
- $P^2 = P \qquad A(A^TA)^{-1} A^T \cdot A(A^TA^{-1})^{\cancel{-1}} \cdot A^T$

Least squares

Fitting by a line $(1,1), (2,2), (3,2)$

$b = Ct + D$



$$\begin{aligned} C + D &= 1 \\ C + 2D &= 2 \\ C + 3D &= 2 \end{aligned} \qquad \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} C \\ D \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$$

$$A \cdot x = b$$